Three Classical Analytic Estimates for Explicit-Formula Coefficients of the Riemann Zeta Function

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Abstract

We formulate three classical analytic estimates for coefficient families arising in explicit-formula expansions indexed by the non-trivial zeros of the Riemann zeta function: (i) a dyadic L^2 tail bound, (ii) a large-sieve–type mean-square inequality for exponential sums over zero ordinates *without* any global spacing hypothesis, and (iii) a uniform mean-square tail bound for a weighted explicit-formula expansion. The arguments rely only on Stirling asymptotics for Γ, the Riemann–von Mangoldt formula (including unit-interval zero counts), and a classical large-sieve inequality for separated exponentials.

1 Introduction

Let $\rho = \beta + i\gamma$ range over the non-trivial zeros of the Riemann zeta function, paired under the functional equation. Write the explicit-formula contribution to $\psi(x)$ in the form

$$\log R(s) = \sum_{\rho} a_\rho \, e^{(\rho-1)s} + \cdots , \qquad j_{\mathrm{rel}}(s) = \sum_{\rho} \frac{\rho-1}{\rho} \, a_\rho \, e^{(\rho-1)s} + \cdots ,$$

and define coefficients

$$b_\rho := \frac{\rho-1}{\rho} \, G(\rho), \qquad G(s) := \pi^{-s/2}\Gamma\Big(\frac{s}{2}\Big) \, H(s).$$

Here $H(s)$ is a holomorphic factor coming from Mellin transforms of test functions, rational weights, and possible ζ'/ζ terms, assumed to be of at most polynomial growth in vertical strips with $\Re(s+\kappa) \geq 1/2 + \varepsilon$ for some $\kappa, \varepsilon > 0$. We present three self-contained analytic estimates for these coefficients and their dyadic aggregates. No reference is made to ERURH or to RH in what follows.

2 Standing hypotheses and preliminaries

Let $\rho = \beta + i\gamma$ range over the non-trivial zeros of $\zeta(s)$, counted with multiplicity, and let

$$b_\rho := \frac{\rho-1}{\rho} \, G(\rho), \qquad G(s) := \pi^{-s/2}\Gamma\Big(\frac{s}{2}\Big) \, H(s).$$

The estimates in this note are stated and proved under the following *explicit* hypotheses, which isolate exactly the analytic input required.

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Standing hypotheses

(H1) (Holomorphic model with gamma factor). The auxiliary factor $H(s)$ is holomorphic in a neighbourhood of the closed strip $0 \le \Re s \le 1$, and the coefficients b_ρ are defined by the above formula (in particular, the completed zeta-factor $\pi^{-s/2}\Gamma(s/2)$ is present in $G(s)$).

(H2) (Polynomial vertical growth of H). There exist constants $C_H > 0$ and $A_H \ge 0$ such that

$$|H(\sigma + it)| \le C_H(1 + |t|)^{A_H} \qquad (0 \le \sigma \le 1,\ t \in \mathbb{R}).$$

(H3) (Riemann–von Mangoldt and unit-interval zero counts). Let $N(T)$ denote the number of non-trivial zeros $\rho = \beta + i\gamma$ with $0 < \beta < 1$ and $0 < \gamma \le T$, counted with multiplicity. We use the classical Riemann–von Mangoldt formula and its consequence that, for $|y| \ge 3$,

$$\#\{\rho :\ y < \Im(\rho) \le y + 1\} \ \ll\ \log(|y| + 2),$$

with an absolute implied constant.

Remark 2.1 (Verifying the hypotheses in explicit-formula settings)**.** Hypothesis **(H1)** is automatic once one works with the completed zeta function $\Lambda(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$, from which the gamma factor enters the explicit formula. Hypothesis **(H2)** holds (with room to spare) when $H(s)$ is a Mellin transform of a smooth compactly supported test function, since such Mellin transforms are entire and rapidly decaying in vertical strips. Rational weights contribute at most polynomial growth.

If one wishes to incorporate additional factors such as $(\zeta'/\zeta)(s+\kappa)$ into $H(s)$, then for the present arguments it suffices that the resulting H remain holomorphic in a neighbourhood of $0 \le \Re s \le 1$ and satisfy **(H2)** there. A simple sufficient condition is $\kappa > 1$, which places $s + \kappa$ in the zero-free half-plane $\Re > 1$ for all s in the strip.

All implied constants may depend on C_H, A_H (and hence on the specific choice of H), but are otherwise absolute.

We record three standard lemmas (Stirling decay, local zero counting, and a classical large-sieve inequality for separated exponentials) that will be used repeatedly.

Lemma 2.2 (Stirling decay for the coefficients)**.** *Assume Hypotheses **(H1)–(H2)**. In particular, $H(s)$ is holomorphic in a neighbourhood of the strip $0 \le \Re s \le 1$ and, for some $A_H \ge 0$,*

$$H(\sigma + it) \ll (1 + |t|)^{A_H} \qquad (0 \le \sigma \le 1,\ t \in \mathbb{R}).$$

Then there exist constants $c > 0$ and $k \ge 0$ (depending only on A_H) such that for every non-trivial zero $\rho = \beta + i\gamma$ of $\zeta(s)$,

$$|b_\rho| \ll e^{-c|\gamma|}\,(1 + |\gamma|)^k, \qquad |b_\rho|^2 \ll e^{-2c|\gamma|}\,(1 + |\gamma|)^{2k}.$$

Proof. Write $s = \sigma + it$ with $0 \le \sigma \le 1$. By Stirling's formula in vertical strips (see, e.g., [1, §2.3]),

$$\left|\Gamma\Big(\frac{s}{2}\Big)\right| \ll (1 + |t|)^{\sigma/2 - 1/2} \exp\Big(-\frac{\pi}{4}|t|\Big),$$

and trivially $|\pi^{-s/2}| = \pi^{-\sigma/2} \asymp 1$ on $0 \le \sigma \le 1$. Hence

$$|G(\sigma + it)| = |\pi^{-s/2}| \cdot \left|\Gamma\Big(\frac{s}{2}\Big)\right| \cdot |H(s)| \ll (1 + |t|)^{A_H + 1} \exp\Big(-\frac{\pi}{4}|t|\Big).$$

For a zero $\rho = \beta + i\gamma$ we have $0 < \beta < 1$, and for $|\gamma| \geq 1$,

$$\Big|\frac{\rho - 1}{\rho}\Big| \leq 1 + \frac{1}{|\rho|} \leq 2.$$

Therefore,

$$|b_\rho| = \Big|\frac{\rho - 1}{\rho}\Big| \cdot |G(\rho)| \ll (1 + |\gamma|)^{A_H + 1} \exp\Big(-\frac{\pi}{4}|\gamma|\Big),$$

which implies the claimed bound with $c = \pi/4$ and $k = A_H + 1$ (after adjusting the implied constant to cover the finitely many zeros with $|\gamma| < 1$). □

Lemma 2.3 (Local zero counting)**.** *Let $N(T)$ be the number of non-trivial zeros $\rho = \beta + i\gamma$ with $0 < \beta < 1$ and $0 < \gamma \leq T$, counted with multiplicity. Then, for $T \geq 3$,*

$$N(T) = \frac{T}{2\pi} \log \frac{T}{2\pi} - \frac{T}{2\pi} + O(\log T),$$

and consequently

$$N(T + 1) - N(T) \ll \log T.$$

In particular, for every $y \geq 3$ the interval $(y, y + 1]$ contains $\ll \log y$ ordinates γ (counted with multiplicity). By symmetry of the zeros under $\gamma \mapsto -\gamma$, the same bound holds for $y \leq -3$, hence for all $|y| \geq 3$,

$$\#\{\rho : \ y < \Im(\rho) \leq y + 1\} \ \ll \ \log(|y| + 2).$$

Proof. This is the classical Riemann–von Mangoldt formula; see [1, Ch. 9]. Taking differences gives

$$N(T + 1) - N(T) = \frac{1}{2\pi} \log T + O(1),$$

which implies $N(T + 1) - N(T) \ll \log T$ for $T \geq 3$. □

Lemma 2.4 (Large sieve for separated exponentials)**.** *Let $\lambda_1, \ldots, \lambda_N \in \mathbb{R}$ satisfy $|\lambda_m - \lambda_n| \geq \delta > 0$ for all $m \neq n$. Then for any complex coefficients $a_1, \ldots, a_N$ and any $L > 0$,*

$$\int_0^L \Big|\sum_{n=1}^N a_n e^{i\lambda_n u}\Big|^2 du \leq \Big(L + \frac{2\pi}{\delta}\Big) \sum_{n=1}^N |a_n|^2.$$

Proof. This is [3, Theorem 1.3]; see also [2, Ch. 7]. □

3 Theorem A: Dyadic L^2 tail bound

Theorem 3.1 (Dyadic L^2 tail bound)**.** *Assume Hypotheses **(H1)–(H3)**. Then there exist constants $c > 0$, $k \geq 0$ and $T_0 \geq 3$ such that, for all $T \geq T_0$,*

$$\sum_{\substack{\rho \\ T < |\gamma| \leq 2T}} |b_\rho|^2 \ \ll \ T^{2k+1} (\log T)\, e^{-2cT}. \tag{1}$$

In particular, there exist constants $C > 0$ and $A > 0$ such that, for all $T \geq T_0$,

$$\sum_{\substack{\rho \\ T < |\gamma| \leq 2T}} |b_\rho|^2 \ \leq \ C\, T (\log T)^A.$$

Proof. By Lemma 2.2, there exist $c > 0$ and $k \geq 0$ such that

$$|b_\rho|^2 \ll e^{-2c|\gamma|}\,(1+|\gamma|)^{2k}$$

for every non-trivial zero $\rho = \beta + i\gamma$. For $T < |\gamma| \leq 2T$ we therefore have the uniform bound

$$|b_\rho|^2 \ll e^{-2cT}\,(1+2T)^{2k} \ll e^{-2cT}\,T^{2k}.$$

Summing over the dyadic band and using the Riemann–von Mangoldt bound $N(2T) - N(T) \ll T\log T$ (which follows from Lemma 2.3) gives

$$\sum_{\substack{\rho \\ T<|\gamma|\leq 2T}} |b_\rho|^2 \ll (N(2T) - N(T))\,e^{-2cT}\,T^{2k} \ll T^{2k+1}(\log T)\,e^{-2cT}.$$

In particular the dyadic L^2 tail is exponentially small in T. Since e^{2cT} dominates any fixed power of T and $\log T$, there exist constants $A > 0$ and $T_0 > 0$ such that for all $T \geq T_0$,

$$T^{2k}(\log T)\,e^{-2cT} \leq (\log T)^A,$$

and hence

$$\sum_{\substack{\rho \\ T<|\gamma|\leq 2T}} |b_\rho|^2 \leq C\,T(\log T)^A$$

for a suitable constant $C > 0$. □

Remark 3.2. The proof in fact yields an exponentially small dyadic tail in T; we record only a polynomial bound because it suffices for applications.

4 Theorem B: A large-sieve inequality for zero ordinates

Fix $L > 0$ and a dyadic band $T < |\gamma| \leq 2T$ with T large. Define the spectral sum

$$F(u) = \sum_{T<|\gamma|\leq 2T} b_\rho\, e^{i\gamma u}.$$

Theorem 4.1 (Large-sieve mean-square inequality for zero ordinates)**.** *Assume Hypothesis **(H3)**. Let $T \geq 3$ and $L \geq 1$. For the dyadic band $T < |\gamma| \leq 2T$ define*

$$F(u) = \sum_{\substack{\rho \\ T<|\gamma|\leq 2T}} b_\rho\, e^{i\gamma u}.$$

Then

$$\frac{1}{L}\int_0^L |F(u)|^2\,du \;\ll\; (\log T) \sum_{\substack{\rho \\ T<|\gamma|\leq 2T}} |b_\rho|^2, \tag{2}$$

with an absolute implied constant. In particular, (2) holds in any "admissible range" $1 \ll L \ll T^\alpha$ with $0 < \alpha < 1$.

Lemma 4.2 (Decomposition into 1-separated subfamilies)**.** *Let Λ be a finite multiset of real numbers with the property that every interval of length 1 contains at most M elements of Λ (counted with multiplicity). Then Λ can be written as a disjoint union*

$$\Lambda = \Lambda_1 \cup \cdots \cup \Lambda_M$$

such that each Λ_j is 1-separated, i.e. $|\lambda - \lambda'| \geq 1$ for all distinct $\lambda, \lambda' \in \Lambda_j$.

Proof. Order the elements of Λ (with multiplicity) as $\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_N$. We construct the subfamilies $\Lambda_1, \ldots, \Lambda_M$ greedily.

Assume $\lambda_1, \ldots, \lambda_{n-1}$ have been assigned so that each Λ_j is 1-separated. Consider λ_n. For each j, the 1-separation implies that Λ_j contains *at most one* element in the interval $(\lambda_n - 1, \lambda_n]$. Hence, if λ_n cannot be placed into Λ_j (i.e. would violate 1-separation), then Λ_j already contains an element of $(\lambda_n - 1, \lambda_n]$.

But $(\lambda_n - 1, \lambda_n]$ contains at most M elements of Λ by hypothesis, so at most $M - 1$ of the sets Λ_j can be "blocked" in this way by previously placed elements. Therefore there exists at least one index j for which Λ_j contains no element of $(\lambda_n - 1, \lambda_n]$, and we may assign λ_n to that Λ_j while preserving 1-separation.

Proceeding inductively assigns all elements to $\Lambda_1, \ldots, \Lambda_M$, as required. □

Proof. Let $\Gamma(T)$ denote the multiset of ordinates $\gamma \in \mathbb{R}$ (counted with multiplicity) of zeros $\rho = \beta + i\gamma$ with $T < |\gamma| \leq 2T$. By Lemma 2.3, every interval of length 1 in the ranges $[T, 2T]$ and $[-2T, -T]$ contains $\ll \log T$ ordinates. Thus there exists an absolute constant $C_0 > 0$ such that

$$M := \sup_{y \in \mathbb{R}} \#\big(\Gamma(T) \cap (y, y+1]\big) \leq C_0 \log T$$

for all sufficiently large T.

Apply Lemma 4.2 to $\Gamma(T)$ to obtain a disjoint union $\Gamma(T) = \Gamma_1 \cup \cdots \cup \Gamma_M$, where each Γ_j is 1-separated. Write accordingly

$$F(u) = \sum_{j=1}^{M} F_j(u), \qquad F_j(u) := \sum_{\substack{\rho \\ \gamma \in \Gamma_j}} b_\rho \, e^{i\gamma u}.$$

By Cauchy–Schwarz,

$$|F(u)|^2 = \Big|\sum_{j=1}^{M} F_j(u)\Big|^2 \leq M \sum_{j=1}^{M} |F_j(u)|^2,$$

and hence

$$\int_0^L |F(u)|^2 \, du \leq M \sum_{j=1}^{M} \int_0^L |F_j(u)|^2 \, du. \tag{3}$$

For each fixed j, the ordinates in Γ_j are 1-separated, so Lemma 2.4 with $\delta = 1$ gives

$$\int_0^L |F_j(u)|^2 \, du \leq (L + 2\pi) \sum_{\substack{\rho \\ \gamma \in \Gamma_j}} |b_\rho|^2.$$

Substituting into (3) and summing over j yields

$$\int_0^L |F(u)|^2 \, du \leq M(L + 2\pi) \sum_{T < |\gamma| \leq 2T} |b_\rho|^2.$$

Dividing by L and using $L \geq 1$ gives

$$\frac{1}{L}\int_0^L |F(u)|^2\,du \leq C\,(\log T) \sum_{T<|\gamma|\leq 2T} |b_\rho|^2,$$

for a suitable absolute constant (absorbing the factor $(1+2\pi/L)$). This is exactly (2). □

Corollary 4.3 (Max-bound version)**.** *Under the assumptions of Theorem 4.1, one has*

$$\frac{1}{L}\int_0^L |F(u)|^2\,du \;\ll\; T\,(\log T)^2 \cdot \max_{T<|\gamma|\leq 2T} |b_\rho|^2.$$

Proof. This follows immediately from Theorem 4.1 and the trivial bound $\sum |b_\rho|^2 \leq (N(2T) - N(T)) \max |b_\rho|^2 \ll T(\log T) \max |b_\rho|^2$. □

5 Theorem C: A uniform RMS tail bound

Let $S \geq 3$ and $L > 0$, and fix a parameter $B > 0$. Define the tail observable

$$\widetilde{F}_{\text{tail}}(s) \;:=\; s^{-2} \sum_{|\gamma|>S^B} b_\rho\, e^{i\gamma s},$$

and the windowed root-mean-square quantity

$$K_{\text{tail}}(S,L)^2 \;:=\; \frac{1}{L}\int_S^{S+L} \big|\widetilde{F}_{\text{tail}}(s)\big|^2\,ds.$$

Theorem 5.1 (Uniform RMS tail bound)**.** *Assume Hypotheses **(H1)**–**(H3)** and fix $B > 0$. Then there exist constants $c' > 0$ and $S_0 \geq 3$ such that, for all $S \geq S_0$ and all $L > 0$,*

$$K_{\text{tail}}(S,L)^2 \;\ll\; S^{-4}\exp\!\big(-2c'S^B\big). \tag{4}$$

In particular, $K_{\text{tail}}(S,L)^2 \leq M_{\text{tail}}^2$ uniformly in S and L, for a suitable constant $M_{\text{tail}} > 0$.

Proof. By Lemma 2.2, there exist $c > 0$ and $k \geq 0$ such that

$$|b_\rho| \ll e^{-c|\gamma|}(1+|\gamma|)^k.$$

We first show that the tail sum $\sum_{|\gamma|>S^B} |b_\rho|$ is exponentially small in S^B. Decompose the region $|\gamma| > S^B$ into dyadic bands $(2^m S^B, 2^{m+1}S^B]$, $m \geq 0$. On each band we have $|b_\rho| \ll e^{-c2^m S^B}(2^m S^B)^k$, and by Lemma 2.3 (equivalently, by Riemann–von Mangoldt),

$$\#\{\rho:\; 2^m S^B < |\gamma| \leq 2^{m+1}S^B\} \;=\; O\big(2^m S^B \log(2^m S^B)\big).$$

Hence

$$\sum_{\substack{\rho \\ 2^m S^B<|\gamma|\leq 2^{m+1}S^B}} |b_\rho| \ll (2^m S^B)^{k+1}\log(2^m S^B)\, e^{-c2^m S^B}.$$

Summing over $m \geq 0$ gives

$$\sum_{\substack{\rho \\ |\gamma|>S^B}} |b_\rho| \ll \sum_{m=0}^{\infty} (2^m S^B)^{k+1}\log(2^m S^B)\, e^{-c2^m S^B} \ll \exp(-c'S^B)$$

for some $c' > 0$ and all $S \geq S_0$, since the exponential factor decays super-geometrically in m.

Now for $s \in [S, S+L]$ we have $|s^{-2}| \leq S^{-2}$, and by absolute convergence,

$$|\widetilde{F}_{\text{tail}}(s)| \leq S^{-2} \sum_{|\gamma|>S^B} |b_\rho| \ll S^{-2} \exp(-c' S^B).$$

Therefore

$$K_{\text{tail}}(S,L)^2 = \frac{1}{L} \int_S^{S+L} |\widetilde{F}_{\text{tail}}(s)|^2 \, ds \ll \frac{1}{L} \int_S^{S+L} S^{-4} e^{-2c'S^B} \, ds \ll S^{-4} e^{-2c'S^B},$$

which is (4). □

Remark 5.2. Because the coefficients b_ρ decay exponentially in $|\gamma|$ under **(H1)**–**(H2)**, the tail bound above is essentially pointwise and does not require any mean-square input beyond absolute convergence. One may also derive comparable bounds by combining dyadic decompositions with Theorems 3.1 and 4.1, but the direct argument above is simpler.

6 Remarks

- The proofs use only classical inputs: Stirling's formula for Γ, the Riemann–von Mangoldt formula (including unit-interval zero counts), and the large sieve for separated exponentials (Lemma 2.4).
- Under **(H1)**–**(H2)**, the dyadic tails in Theorem 3.1 are in fact exponentially small in the dyadic parameter.
- The large-sieve inequality in Theorem 4.1 does not assume any uniform lower bound on gaps between ordinates; it uses only the local counting information encoded in **(H3)**.
- Variants with different test functions or normalizations of b_ρ can be handled similarly by modifying $H(s)$ and verifying **(H1)**–**(H2)** for the resulting factor.

References

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